

Artist-Conditional Lyric Generation with QLoRA Adapter Blending

CENG 467 — Natural Language Understanding & Generation

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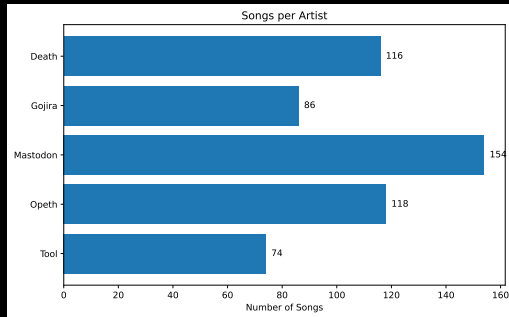
IYTE — Instructor: Prof. Aytuğ Onan

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Task: five same-genre metal artists

Artist	Songs	Style
Mastodon	154	sludge / prog
Death	116	technical / philos.
Opeth	118	poetic / melancholic
Gojira	86	environmental
Tool	74	cryptic / progressive

Same genre on purpose: learn *artist*, not genre.



Method

- ▶ Base: **Gemma 3n E4B** ($\sim 4\text{B}$), QLoRA 4-bit NF4
- ▶ One **LoRA r8** adapter per artist (all linear layers)
- ▶ **Blend** two adapters in delta space (rank concatenation):

$$A_{\text{blend}} = \begin{bmatrix} \alpha A_1 \\ (1 - \alpha) A_2 \end{bmatrix}, \quad B_{\text{blend}} = \begin{bmatrix} B_1 & B_2 \end{bmatrix}$$

α = weight of artist 1; endpoints recover the pure adapters exactly.

Why fine-tune? The base model can't be prompted

Base, zero-shot

[User 0001]

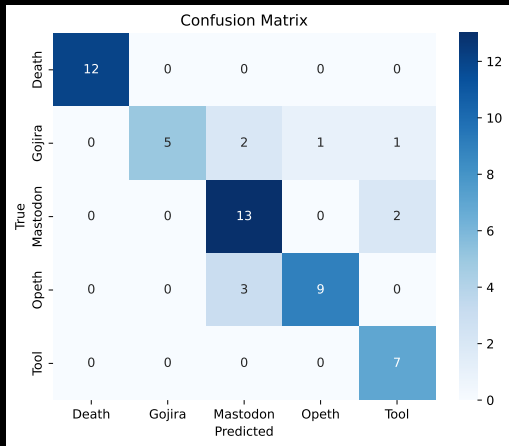
This is my band/artist page is where you should enter your range of audio, video, preset unavailable for your region... to read lyrics from web. If you followed any links on this page...

Base → web/HTML junk. Adapter → clean artist lyrics.

Gojira adapter

*I took a boat across the ocean
And I sail away, sailing home
On an open sea that is so deep
I'm falling into my dreams*

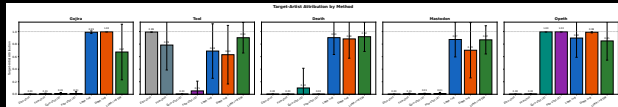
The judge: an attribution classifier



- ▶ RoBERTa-base, accuracy **0.873**
- ▶ Strong in-domain (Death f1 1.00)
- ▶ No “other” class $\Rightarrow$ out-of-distribution input is forced onto a **sink** class

Attribution is a *sanity check*, valid only on real lyric input.

Adapters encode style; baselines collapse to sinks

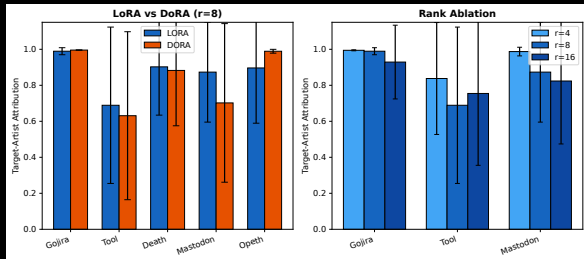


Plain LoRA r8

Target	Attribution
Gojira	0.989
Death	0.903
Opeth	0.896
Mastodon	0.873
Tool	0.689
base → Tool	≈0.99
instruct → Opeth	≈0.99

Adapters attribute with no artist name in the prompt; both baseline families collapse to a sink class.

Ablation: LoRA rank



Rank	Gojira	Tool	Mastodon
r4	.994	.838	.987
r8	.989	.689	.873
r16	.929	.755	.824

r16 consistently overfits; r4 matches/beats r8 on attribution but can lose on perplexity — rank choice depends on the metric.

Style-weighted loss exposes the judge

Plain Tool — score **0.003**

*You know better than me
What's what and why
I might not be a prodigy
But I can write a rhyme*

SW Tool — score **0.996**

*Yeah, yeah
Yeah, hey, hey
Hey, hey, hey, hey
Hey, hey, hey, hey (×10)*

	Gojira (plain→SW)	Tool (plain→SW)
Attribution	.989 → .674	.689 → .901
Types/sample	1.6 → 5.7 (3.6×)	2.5 → 3.3 (1.3×)
Occ./type	1.2 → 2.1	1.2 → 8.6

Higher score, worse lyrics: Tool's gain is pure repetition, not breadth.

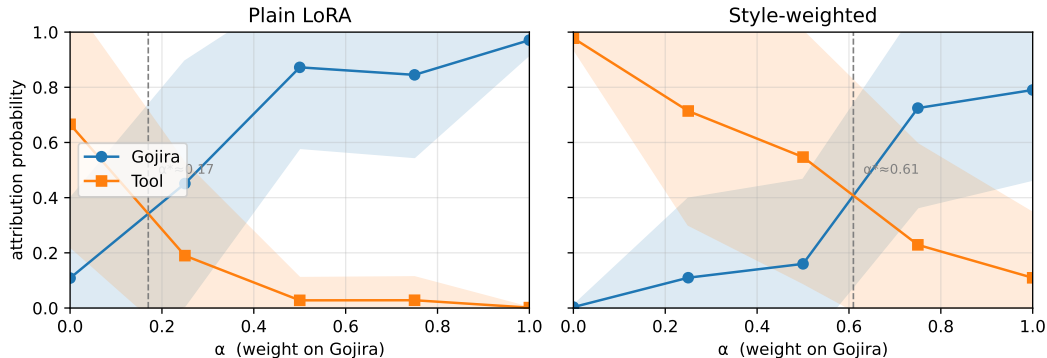
Four metrics, five artists — they disagree

Artist	Attribution (plain→SW)	Types/sample (plain→SW)	Occ./sample (plain→SW)	distinct-2 (plain→SW)
Gojira	.99 → .67	1.6 → 5.7	1.9 → 11.9	.96 → .71
Tool	.69 → .90	2.5 → 3.3	3.1 → 28.5	.88 → .74
Death	.90 → .92	1.7 → 1.5	2.9 → 7.2	.67 → .63
Mastodon	.87 → .87	2.0 → 3.3	2.7 → 4.9	.90 → .70
Opeth	.90 → .85	0.7 → 2.8	0.7 → 7.1	.92 → .71

Distinctive-token *breadth* (types) rises for Gojira/Opeth/Mastodon but not Tool/Death; *occurrences* inflate everywhere (repetition); distinct-2 falls for all (degeneration); attribution moves both ways. No single number captures style.

Blending: 50/50 weights $\neq$ 50/50 style

Gojira-Tool blend: plain vs style-weighted



Crossover moves from $\alpha=0.17$ (plain) to 0.61 (SW): the loss flips which artist dominates the blend.

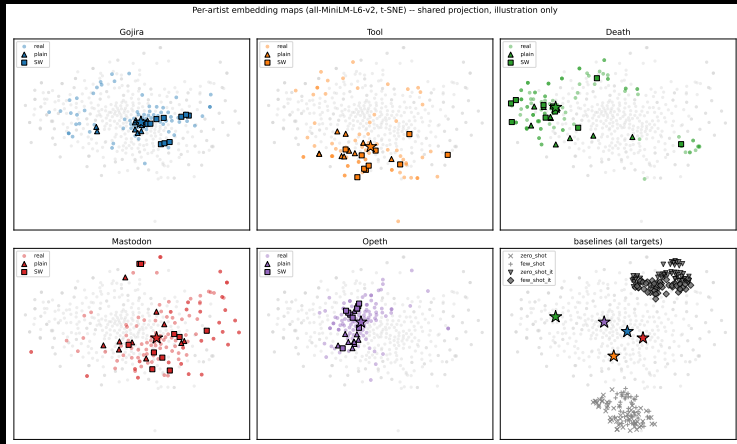
A blend, generated: 50/50 Gojira + Death

*Enter the land of darkness
An ancient race now destroyed
Lost in time and space forever
Never to return*

*In the sky so clear you can see
The flames burning bright below
A giant dragon is breathing fire
It grows ever stronger with each blow
Feel the heat flowing through your veins
Fire at will, breathe and destroy
Dragon is approaching fast
Breath comes out as red hot steel
Leave a trail of death behind you*

$\alpha=0.5$ Gojira-SW $\times$ Death-SW. The interpolated style reads as a *third* artist — Mastodon 0.79 (Gojira 0.12, Death 0.09).

Independent instrument, same verdict



Adapters land on each artist's real-song cluster; base and instruct baselines sit in two separate out-of-distribution islands. Three axes disagree (judge / breadth / repetition); faithful reproduction of an offensive source is a deployment risk.

Style is not one number.